

Measurement Practice Workbook

Define the question. Compare like with like. Preserve the original reading.

This workbook gives you two worked examples, three practice questions and an original blank record page. It is intended for learners practising arithmetic used in sewing measurement notes.

Scope: Every numerical example is invented. The worksheet does not report fabric, printer or classroom experiments. You can practise without claiming that an observation was made.

What you will practise

- Choose the correct denominator for stretch and recovery.
- Calculate dimensional and rectangular-area changes separately.
- Convert length units while keeping the original inputs.
- Explain why different axis ratios cannot share one exact uniform correction.

Unit reference

From	To	Operation
centimetres	millimetres	Multiply by 10
millimetres	centimetres	Divide by 10
millimetres	inches	Divide by 25.4

The inch conversion is exact: 1 inch = 25.4 mm. Extra decimal digits in a conversion describe arithmetic, not the precision of an observation.

Before a real observation

Name the exact distance, its unit, the reference state and the measurement method. Record conditions and instrument resolution honestly. Keep raw readings separate from display rounding.

Stretch and Recovery Use Different Denominators

Invented readings for one labelled distance: rest 120 mm, extended 144 mm, released 123 mm. These are three states of a hypothetical example, not measurements of a tested fabric.

State	Hypothetical reading	Meaning in this example
Rest	120 mm	Reference before extension
Extended	144 mm	Reading while extended
Released	123 mm	Reading after release

1. Stretch relative to the rest length

Stretch = (extended - rest) / rest x 100
 $(144-120)/120 \times 100 = 20\%$

The 24 mm extension is compared with the 120 mm rest length.

2. Extension still present after release

Remaining extension = (released - rest) / rest x 100
 $(123-120)/120 \times 100 = 2.5\%$

The 3 mm remaining extension is again compared with the rest length.

3. Recovery of the introduced extension

Recovery = (extended - released) / (extended - rest) x 100
 $(144-123)/(144-120) \times 100 = 21/24 \times 100 = 87.5\%$

Do not subtract 2.5% from 100% to obtain this recovery. The residual percentage uses the 120 mm rest length. Recovery here uses the 24 mm introduced extension. Different denominators answer different questions.

A real recovery statement would also need a defined release time, loading method and observation conditions. This arithmetic does not establish a fabric's performance.

Two Side Changes Produce One Area Product

Invented rectangular readings: before 40 cm wide and 30 cm long; after 38.8 cm wide and 29.4 cm long. Keep the same units and named directions throughout.

Quantity	Before	After	Signed change
Width	40 cm	38.8 cm	-3%
Length	30 cm	29.4 cm	-2%
Rectangular area	1200 cm ²	1140.72 cm ²	-4.94%

Keep the sign convention explicit

Signed dimensional change = (after - before) / before x 100.
 A negative result means a smaller reading. If contraction is expressed as a positive reduction, these examples have 3% width contraction and 2% length contraction.

Calculate the area from the sides

Before area = 40 x 30 = 1200 cm².
 After area = 38.8 x 29.4 = 1140.72 cm².
 Area ratio = 1140.72/1200 = 0.9506.
Area contraction = (1-0.9506) x 100 = 4.94%.

The side ratios multiply: 0.97 x 0.98 = 0.9506. Adding 3% and 2% gives 5%, which is not the exact area contraction. Area alone does not reveal which direction changed more.

Check your interpretation

These calculations compare rectangular readings. They do not prove uniform behaviour across a whole cloth, identify a care-process cause or calculate a suitable garment allowance. Those claims would require additional evidence.

Three Practice Questions

Use the invented values below. Show the formula, unit and reference state. Keep working precision until the final displayed result. Answers are on page 6.

A. Convert a retained input

A hypothetical record contains 250 mm. Express it in centimetres and inches. Show the inch result to four decimal places. Keep 250 mm as the original input.

Working	Answer and unit

B. Keep both print axes

A reference target is 80 x 80 mm. Invented readings are 78.4 mm wide and 79.2 mm high. Find each signed error percentage and the multiplier needed for each axis. Can one multiplier correct both exactly in a uniform model?

Axis	Signed error	Required multiplier
Width		
Height		

C. Compare rectangular area

Invented before dimensions are 25 x 20 cm. After dimensions are 24 x 19.2 cm. Calculate before area, after area and positive area contraction. Explain why the result differs from adding the two side-contraction percentages.

Before area	After area	Area contraction

An Original Measurement Record

Use this page for a labelled practice scenario or for your own observations. Circle one: **HYPOTHETICAL / OBSERVED**. Do not present invented readings as a physical test.

Record label / date	Dimension or axis	Unit

State or reference	Raw reading	Instrument resolution / note

Method and conditions

Exact edges, timing, loading or care conditions; instrument; file/settings when relevant

Calculation and rounding

Question and denominator	Calculation / retained working value	Displayed answer and rounding

Keep your raw reading. A calculated percentage and a rounded display are separate from the observation. This blank record is a worksheet, not evidence that any experiment has been performed.

Check the Reasoning, Then the Result

A. 250 mm

$250/10 = 25$ cm.

$250/25.4 = 1250/127$ inches, approximately **9.8425 in** to four decimal places. Retain 250 mm as the original value; the rounded inch display is not a new observation.

B. 80 mm target, 78.4 mm width, 79.2 mm height

Width error: $(78.4-80)/80 \times 100 = -2\%$.

Height error: $(79.2-80)/80 \times 100 = -1\%$.

Width multiplier: $80/78.4 = 50/49$, approximately **1.020408**.

Height multiplier: $80/79.2 = 100/99$, approximately **1.010101**.

The multipliers differ, so one uniform multiplier cannot make both axes exactly 80 mm.

C. 25 x 20 cm becomes 24 x 19.2 cm

Before area = $25 \times 20 = 500$ cm².

After area = $24 \times 19.2 = 460.8$ cm².

Area contraction = $(1-460.8/500) \times 100 = 7.84\%$.

Each side contracts 4%, but $0.96 \times 0.96 = 0.9216$. Adding 4% + 4% gives 8%, which is not the exact product result.

Further explanation

[Fabric Measurement Guide: Stretch, Recovery and Care](#) explains comparison definitions, denominators and retained inputs.

Publication and scope: Prepared by Sewlore with AI-assisted writing. Numerical examples are hypothetical; no physical testing was performed. Edition 9 October 2026. All rights reserved for any applicable copyright. No new open reuse licence is granted.